Alex Shaffer, Istiaque Ahmed
Email: aws9, miahmed2@illinois.edu
CS 555: Homework # 2
Spring 2026
GitHub Repository: `https://github.com/redhorn144/CS555`

Problem 1
optional: summarize the problem

Solution:

(a) For this part we have implemented a simple the point finite difference stencil to solve Burgers' equation. Plots of the solution for both Chebyshev and uniform nodes are shown in Figures 1 and 2, respectively. For both of these N is taken to be 200, and the time step is taken to be smaller than the CFL constrint from part c. Stapshots of the solution are taken at for $t = 0.1k, \quad 0 \leq k \leq 20$. In these particular cases we can see that the Chebyshev nodes do a better job at resolving the shock in the boundary layer. This is because the Chebyshev nodes are clustered more closely together near the boundaries. To be consistent with the paper we take $\nu = 1/100\pi$, but to be consistent with the homework handout we take $u^0 = \sin(\pi x)$ instead of $u^0 = -\sin(\pi x)$.

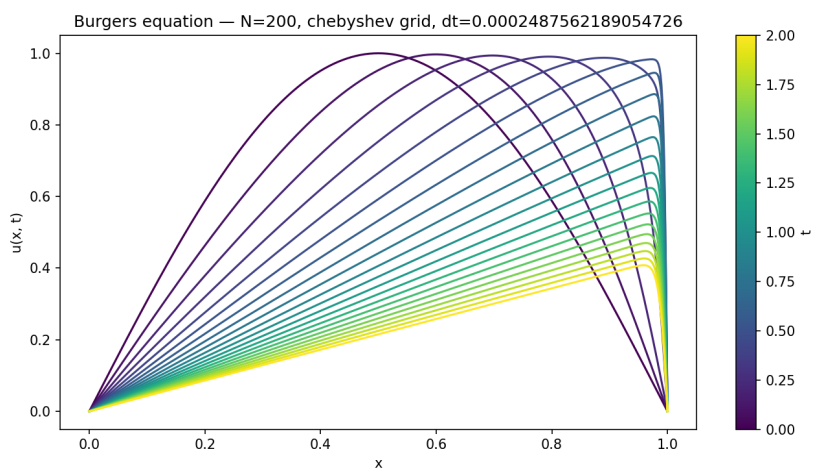

Figure 1: Solutions to Burgers' equation with Chebyshev nodes.

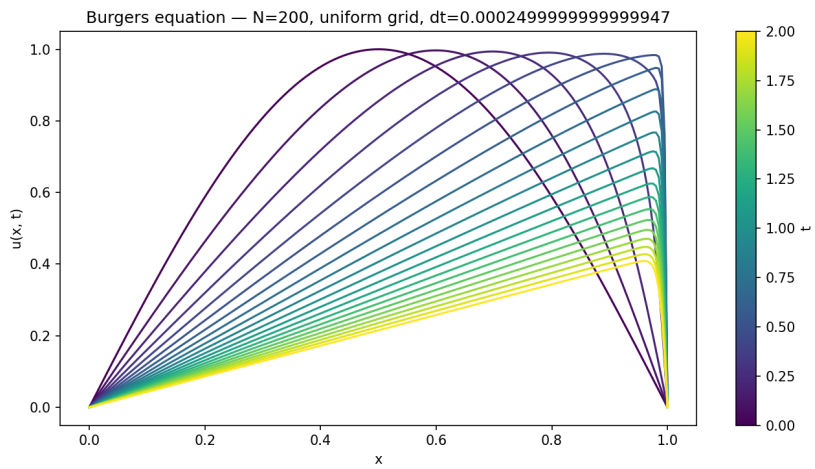

Figure 2: Solutions to Burgers' equation with uniform nodes.

(b) Now we are asked to calculate the maximum value of the derivative of u $s(t) = max_x|u_x(x,t)|$. This is done with N = 200 again for both the uniform nodes and the Chebyshev nodes. The plot of $s(t)$ as well as $s^* = max_t s(t)$ is shown in Figures 3 and 4 for the Chebyshev and uniform nodes, respectively. The paper scales $t* = argmax s(t)$ by π. But without that scaling the analytical values of s^* and t^* are $s^* = 152.00516$ and $t^* = 0.51047$. The Chebyshev node case does notably better for determining s^* and t^*.

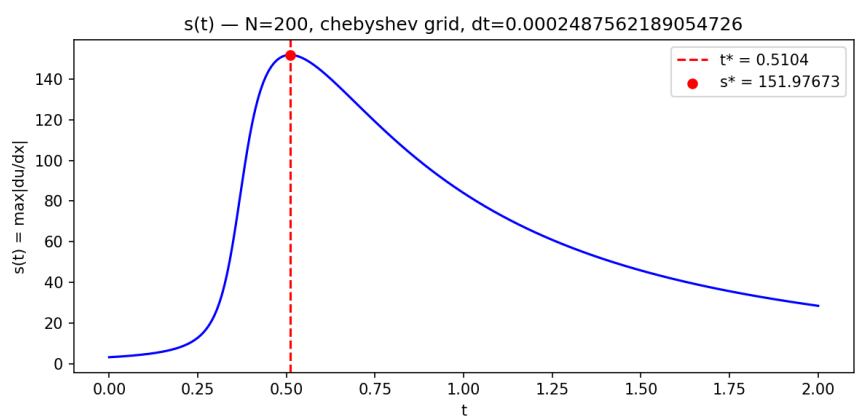

Figure 3: Your descriptive caption here.

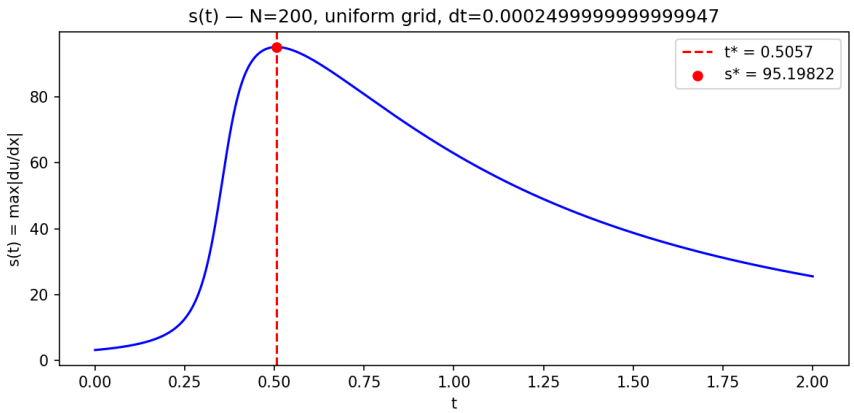

Figure 4: Your descriptive caption here.

(c)

N	dt	nsteps	s^*	rel_err
25	3.864938e-03	518	72.284303	5.245e-01
50	1.960784e-03	1020	142.296243	6.387e-02
100	9.900990e-04	2020	151.398113	3.994e-03
200	4.975124e-04	4021	151.977056	1.849e-04
400	2.493766e-04	8021	152.005901	4.878e-06
800	1.248439e-04	16020	152.005832	4.419e-06
1500	6.662225e-05	30020	152.005385	1.481e-06
3000	3.332223e-05	60020	152.005220	3.941e-07
6000	1.666389e-05	120020	152.005176	1.073e-07
10000	9.999000e-06	200020	152.005167	4.552e-08
15000	6.666222e-06	300020	152.005164	2.601e-08

Table 1: Convergence data for various N.

(d)

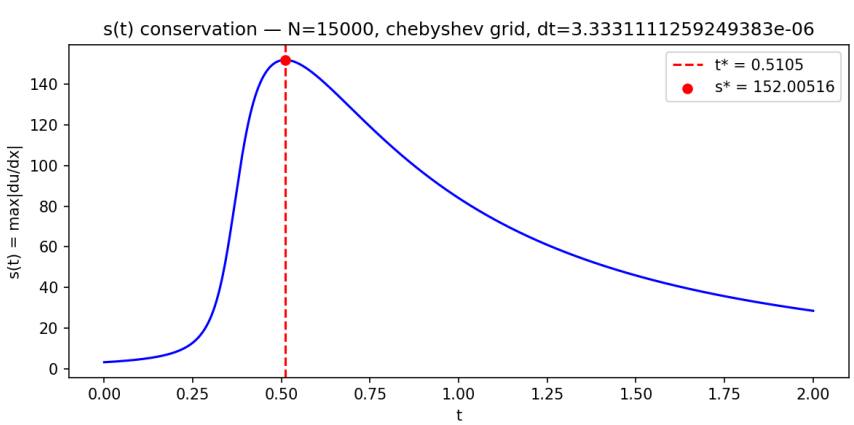

Figure 5: Conservation of $s_{\max}$ using Chebyshev nodes.

(e)